CHAPTER EXERCISE: Answers for this chapter start on page 319.

A calculator should NOT be used on the following questions.

1

For $i = \sqrt{-1}$, which of the following is equivalent to $(5 - 3i) - (-2 + 5i)$?

A) $3 - 8i$
B) $3 + 2i$
C) $7 - 8i$
D) $7 + 2i$

2

Given that $i = \sqrt{-1}$, which of the following is equal to $i(i + 1)$?

A) $i - 2$
B) $i - 1$
C) $i + 1$
D) 0

3

$$i^4 + 3i^2 + 2$$

Which of the following is equal to the expression above? (Note: $i = \sqrt{-1}$)

A) i
B) -1
C) 0
D) 1

4

$$2 + 3i + 4i^2 + 5i^3 + 6i^4$$

If the expression above is equivalent to $a + bi$, where a and b are constants, what is the value of $a + b$? (Note $i = \sqrt{-1}$)

A) 2
B) 6
C) 10
D) 12

5

$$(6 + 2i)(2 + 5i)$$

If the expression above is equivalent to $a + bi$, where a and b are constants, what is the value of a ?

A) 2
B) 12
C) 22
D) 34

6

Which of the following is equal to $3(i + 2) - 2(5 - 4i)$? (Note: $i = \sqrt{-1}$)

A) $16 - 5i$
B) $-4 + 7i$
C) $-4 + 11i$
D) $16 + 11i$

7

For $i = \sqrt{-1}$, which of the following is equivalent to $3i(i + 2) - i(i - 1)$?

A) $-4 + 7i$
B) $-2 + 7i$
C) $-4 + 5i$
D) $-2 + 5i$

8

For $i = \sqrt{-1}$, which of the following is equal to i^{93} ?

A) -1
B) 1
C) $-i$
D) i

9

Which of the following complex numbers is equivalent to $(3-i)^2$? (Note: $i=\sqrt{-1}$)

A) $8-6i$

B) $8+6i$

C) $10-6i$

D) $10+6i$

10

$$(-i)^2-(-i)^4$$

In the complex number system, what is the value of the expression above? (Note: $i=\sqrt{-1}$)

A) -2

B) 0

C) 1

D) 2

11

$$(5-2i)(4-3i)$$

Which of the following is equal to the expression above? (Note: $i=\sqrt{-1}$)

A) $14-7i$

B) $14-23i$

C) $26+7i$

D) $26-23i$

12

Which of the following is equal to $\frac{1}{i}+\frac{1}{i^2}+\frac{1}{i^4}$?

(Note: $i=\sqrt{-1}$)

A) $-i$

B) i

C) 0

D) 1

13

Which of the following is equal to $\frac{1-3i}{3+i}$?

(Note: $i=\sqrt{-1}$)

A) $-i$

B) i

C) $-\frac{5}{4}i$

D) $\frac{3}{4}-\frac{5}{4}i$

14

Which of the following complex numbers is equivalent to $\frac{2-i}{2+i}$? (Note: $i=\sqrt{-1}$)

A) $\frac{3}{5}-\frac{4}{5}i$

B) $1-\frac{4}{5}i$

C) $\frac{5}{3}-\frac{4}{3}i$

D) $1-\frac{4}{3}i$

15

$$\frac{4+i}{1-i}+\frac{2-i}{1+i}$$

Which of the following is equal to the expression above? (Note: $i=\sqrt{-1}$)

A) $-2-i$

B) $2+i$

C) $4+i$

D) $4-i$

Chapter 19: Complex Numbers

CHAPTER EXERCISE:

1. [C] $(5-3i)-(-2+5i) = 5-3i+2-5i = 7-8i$

2. [B] $i(i+1) = i^2+i = -1+i$

3. [C] $i^4+3i^2+2 = 1-3+2 = 0$

4. [A] $2+3i+4i^2+5i^3+6i^4 = 2+3i+4(-1)+5(-i)+6(1) = 4-2i$
   So $a=4$, $b=-2$, and $a+b=2$.

5. [A] $(6+2i)(2+5i) = 12+30i+4i+10i^2 = 12+34i+10(-1) = 2+34i$
   Therefore, $a=2$.

6. [C] $3(i+2)-2(5-4i) = 3i+6-10+8i = -4+11i$

7. [B] $3i(i+2)-i(i-1) = 3i^2+6i-i^2+i = -3+6i-(-1)+i = -2+7i$

8. [D] $i^{93} = (i^4)^{23}\cdot i = (1)^{23}\cdot i = i$

9. [A] $(3-i)^2 = 3^2-6i+i^2 = 9-6i-1 = 8-6i$

10. [A] Deal with the exponents first: $(-i)^2-(-i)^4 = i^2-i^4 = -1-1 = -2$

11. [B] $(5-2i)(4-3i) = 20-15i-8i+6i^2 = 20-23i-6 = 14-23i$

12. [A] $\frac{1}{i}+\frac{1}{i^2}+\frac{1}{i^4} = \frac{1}{i}-1+1 = \frac{1}{i}$

    Now multiply both top and bottom by i to get $\frac{1}{i}\cdot\frac{i}{i} = \frac{i}{i^2} = -i$

13. [A] $\frac{(1-3i)}{(3+i)}\cdot\frac{(3-i)}{(3-i)} = \frac{3-i-9i+3i^2}{9-3i+3i-i^2} = \frac{3-10i+3i^2}{9-i^2} = \frac{3-10i-3}{9-(-1)} = \frac{-10i}{10} = -i$

14. [A] $\frac{(2-i)}{(2+i)}\cdot\frac{(2-i)}{(2-i)} = \frac{2^2-2i-2i+i^2}{4-2i+2i-i^2} = \frac{4-4i+i^2}{4-i^2} = \frac{4-4i-1}{4-(-1)} = \frac{3-4i}{5} = \frac{3}{5}-\frac{4}{5}i$

15. [B] The common denominator is $(1-i)(1+i)$.

$$\frac{(4+i)(1+i)}{(1-i)(1+i)}+\frac{(2-i)(1-i)}{(1-i)(1+i)} = \frac{(4+i+4i+i^2)+(2-2i-i+i^2)}{1+i-i-i^2} = \frac{4+5i-1+2-3i-1}{1-(-1)} = \frac{4+2i}{2}$$
$$= 2+i$$